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Critical Thinking and Study Skills (#1700370)

Version: 2024 - And Beyond (current)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- [ELA.9.C.1.3](#): Write to argue a position, supporting claims using logical reasoning and credible evidence from multiple sources, rebutting counterclaims with relevant evidence, using a logical organizational structure, elaboration, purposeful transitions, and a tone appropriate to the task.

Clarifications:

Clarification 1: See [Writing Types](#) and [Elaborative Techniques](#).

- [ELA.9.C.1.4](#): Write expository texts to explain and analyze information from multiple sources, using a logical organization, varied purposeful transitions, and a tone appropriate to the task.

Clarifications:

Clarification 1: See [Writing Types](#).

- [ELA.9.C.1.5](#): Improve writing by considering feedback from adults, peers, and/or online editing tools, revising for clarity and cohesiveness.
- [ELA.9.C.2.1](#): Present information orally, with a logical organization and coherent focus, with credible evidence, creating a clear perspective.

Clarifications:

Clarification 1: At this grade level, the emphasis is on the content, but students are still expected to follow earlier expectations: volume, pronunciation, and pacing. A clear perspective is the through-line that unites the elements of the presentation.

Clarification 2: For further guidance, see the [Secondary Oral Communication Rubric](#).

- [ELA.9.C.3.1:](#) Follow the rules of standard English grammar, punctuation, capitalization, and spelling appropriate to grade level.

Clarifications:

Clarification 1: Skills to be implemented but not yet mastered are as follows:

- Add variety to writing or presentations by using parallel structure and various types of phrases and clauses.
- Use knowledge of usage rules to create flow in writing and presenting.

Clarification 2: See [Convention Progression by Grade Level](#).

- [ELA.9.C.4.1:](#)

Conduct research to answer a question, drawing on multiple reliable and valid sources and refining the scope of the question to align with findings.

Clarifications:

Clarification 1: There is no requirement that students research the additional questions generated.

- [ELA.9.R.2.1:](#) Analyze how multiple text structures and/or features convey a purpose and/or meaning in texts.

Clarifications:

Clarification 1: Students will analyze the use of the following structures: description, problem/solution, chronological, compare and contrast, cause and effect, and sequence.

Clarification 2: Students will evaluate the use of the following features: table of contents, headings, captions, photographs, graphs, charts, illustrations, glossary, footnotes, annotations, and appendix.

- [ELA.9.R.2.2:](#) Evaluate the support an author uses to develop the central idea(s) throughout a text.

Clarifications:

Clarification 1: In this grade level, students are using and responsible for the appeals of logos, ethos, and pathos.

Clarification 2: See [Rhetorical Appeals](#) and [Rhetorical Devices](#).

- [ELA.9.R.2.3:](#) Analyze how an author establishes and achieves purpose(s) through rhetorical appeals and/or figurative language.

Clarifications:

Clarification 1: Figurative language use that students will analyze are metaphor, simile, alliteration, onomatopoeia, personification, hyperbole, meiosis (understatement), allusion, and idiom. Other examples can be used in instruction.

Clarification 2: Students will explain the appropriateness of appeals in achieving a purpose. In this grade level, students are using and responsible for the appeals of logos, ethos, and pathos.

Clarification 3: See [Secondary Figurative Language](#).

Clarification 4: See [Rhetorical Appeals](#) and [Rhetorical Devices](#).

- [ELA.9.R.2.4:](#) Compare the development of two opposing arguments on the same topic, evaluating the effectiveness and validity of the claims.

Clarifications:

Clarification 1: Validity refers to the soundness of the arguments.

- [ELA.9.V.1.1:](#) Integrate academic vocabulary appropriate to grade level in speaking and writing.

Clarifications:

Clarification 1: To integrate vocabulary, students will apply the vocabulary they have learned to authentic speaking and writing tasks independently. This use should be intentional, beyond responding to a prompt to use a word in a sentence.

Clarification 2: Academic vocabulary appropriate to grade level refers to words that are likely to appear across subject areas for the current grade level and beyond, vital to comprehension, critical for academic discussions and writing, and usually require explicit instruction.

- [ELA.9.V.1.2:](#) Apply knowledge of etymology and derivations to determine meanings of words and phrases in grade-level content.

Clarifications:

Clarification 1: Etymology refers to the study of word origins and the ways that words have changed over time.

Clarification 2: Derivation refers to making new words from an existing word by adding affixes.

- [ELA.9.V.1.3:](#) Apply knowledge of context clues, figurative language, word relationships, reference materials, and/or background knowledge to determine the connotative and denotative meaning of words and phrases, appropriate to grade level.

Clarifications:

Clarification 1: Review of words learned in this way is critical to building background knowledge and related vocabulary.

Clarification 2: See [Context Clues](#) and [Word Relationships](#).

Clarification 3: See [9.R.3.1](#) and [Secondary Figurative Language](#).

- [ELA.K12.EE.1.1:](#) Cite evidence to explain and justify reasoning.

Clarifications:

K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- [ELA.K12.EE.2.1:](#) Read and comprehend grade-level complex texts proficiently.

Clarifications:

See [Text Complexity](#) for grade-level complexity bands and a text complexity rubric.

- [ELA.K12.EE.3.1](#): Make inferences to support comprehension.

Clarifications:

Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like “Why is the girl smiling?” or make predictions about what will happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- [ELA.K12.EE.4.1](#): Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

Clarifications:

In kindergarten, students learn to listen to one another respectfully.

In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think _____ because _____.” The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- [ELA.K12.EE.5.1](#): Use the accepted rules governing a specific format to create quality work.

Clarifications:

Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- [ELA.K12.EE.6.1](#): Use appropriate voice and tone when speaking or writing.

Clarifications:

In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd

grade and beyond, students practice appropriate social and academic language to discuss texts.

- **MA.K12.MTR.1.1: Actively participate in effortful learning both individually and collectively.**

Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

- **MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.**

Mathematicians who demonstrate understanding by representing problems in multiple ways:

- Build understanding through modeling and using manipulatives.
- Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
- Progress from modeling problems with objects and drawings to using algorithms and equations.

- Express connections between concepts and representations.
- Choose a representation based on the given context or purpose.

Clarifications:

Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.
- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.

- **MA.K12.MTR.3.1: Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:

Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- **MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others.**

Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

- Communicate mathematical ideas, vocabulary and methods effectively.
- Analyze the mathematical thinking of others.
- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.
- Construct possible arguments based on evidence.

Clarifications:

Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

• **MA.K12.MTR.5.1: Use patterns and structure to help understand and connect mathematical concepts.**

Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

- Focus on relevant details within a problem.
- Create plans and procedures to logically order events, steps or ideas to solve problems.
- Decompose a complex problem into manageable parts.
- Relate previously learned concepts to new concepts.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.

Clarifications:

Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

- Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
- Support students to develop generalizations based on the similarities found among problems.
- Provide opportunities for students to create plans and procedures to solve problems.
- Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- **MA.K12.MTR.6.1: Assess the reasonableness of solutions.**

Mathematicians who assess the reasonableness of solutions:

- Estimate to discover possible solutions.
- Use benchmark quantities to determine if a solution makes sense.
- Check calculations when solving problems.
- Verify possible solutions by explaining the methods used.
- Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to assess the reasonableness of solutions:

- Have students estimate or predict solutions prior to solving.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"
- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justifications.

- **MA.K12.MTR.7.1: Apply mathematics to real-world contexts.**

Mathematicians who apply mathematics to real-world contexts:

- Connect mathematical concepts to everyday experiences.
- Use models and methods to understand, represent and solve problems.

- Perform investigations to gather data or determine if a method is appropriate.
- Redesign models and methods to improve accuracy or efficiency.

Clarifications:

Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
 - Challenge students to question the accuracy of their models and methods.
 - Support students as they validate conclusions by comparing them to the given situation.
 - Indicate how various concepts can be applied to other disciplines.
- [SS.912.P.11.3:](#) Discuss strategies for improving the encoding of memory.
 - [SS.912.P.11.7:](#) Discuss strategies for improving the storage of memories.
 - [SS.912.P.11.12:](#) Discuss strategies for improving the retrieval of memories.
 - [SS.912.P.12.2:](#) Define processes involved in problem solving and decision making.
 - [ELD.K12.ELL.SI.1:](#) English language learners communicate for social and instructional purposes within the school setting.

Course Information

General Notes

This course is designed to develop skills related to critical thinking, learning and problem solving, enabling students to enhance their performance in both academic and non-academic areas. Strategies for acquiring, storing and retrieving information, time management and organizational skills, critical thinking operations and processes, strategies for oral and written communication, and problem solving skills including test taking skills are an integral part of this course.

Florida's Benchmarks for Excellent Student Thinking (B.E.S.T.) Standards

This course includes Florida's B.E.S.T. ELA Expectations (EE) and Mathematical Thinking and Reasoning Standards (MTRs) for students. Florida educators should intentionally embed these standards within the content and their instruction as applicable. For guidance on the implementation of the EEs and MTRs, please visit https://www.cpalms.org/Standards/BEST_Standards.aspx and select the appropriate B.E.S.T. Standards package.

English Language Development ELD Standards Special Notes Section:

Teachers are required to provide listening, speaking, reading and writing instruction that allows English language learners (ELL) to communicate for social and instructional purposes within the school setting. For the given level of English language proficiency and with visual, graphic, or interactive support, students will interact with grade level words, expressions, sentences and discourse to process or produce language necessary for academic success. The ELD standard should specify a relevant content area concept or topic of study chosen by curriculum developers and teachers which maximizes an ELL's need for communication and social skills. To access an ELL supporting document which delineates performance definitions and descriptors, please click on the following link:

https://cpalmsmediaproduct.blob.core.windows.net/uploads/docs/standard_s/eld/si.pdf

General Information

Course Number: 1700370

Course Path: **Section:** Grades PreK to 12 Education Courses >

Grade Group: Grades 9 to 12 and Adult Education Courses >

Subject: Research and Critical Thinking > **SubSubject:** General >

Abbreviated Title: CRIT THINK ST SKLS

Number of Credits: Half credit (.5)

Course Length: Semester (S)

Course Type: Elective Course

Course Level: 2

Course Status: State Board Approved

Grade Level(s): 9,10,11,12

Educator Certifications

One of these educator certification options is required to teach this course.

- Classical Education - Restricted (Elementary and Secondary Grades K-12)

Section 1012.55(5), F.S., authorizes the issuance of a classical education teaching certificate, upon the request of a classical school, to any applicant who fulfills the requirements of s. 1012.56(2)(a)-(f) and (11), F.S., and Rule 6A-4.004, F.A.C. Classical schools must meet the requirements outlined in s. 1012.55(5), F.S., and be listed in the FLDOE Master School ID database, to request a restricted classical education teaching certificate on behalf of an applicant.

Qualifications

As well as any certification requirements listed on the course description, the following qualifications may also be acceptable for the course:

Any academic coverage (any coverage classified as an academic coverage in Rules 6A-4.0101 through 6A-4.0343, Florida Administrative Code).